

## Technical Note

### GLASS SURFACE COATINGS

#### DESCRIPTION

Dorf Ketal Tyzor® organic titanates and zirconates are used to prevent glass surfaces from scratching and to provide adhesion and bonding of other metal oxide finishes to glass and other surfaces with similar characteristics.

When a glass surface is treated at a temperature between its strain and deforming temperatures with the vapor of an organic titanate ester or chelate, the titanate pyrolyzes to form a (TiO<sub>2</sub>)X coating on the glass surface.

#### APPLICATIONS

A thin (<100 nm) film of (TiO<sub>2</sub>)X is virtually transparent. These films are believed to be bound to the glass surface by Ti-O-Si bonds. After a lubricant is applied, these bonds impart considerable scratch resistance to the glass and consequently greatly reduce its fragility.

Tyzor® TPT is generally preferred for glass treatment, but chelates such as Tyzor® AA-75, TE, and PITA also have been used. They can be applied undiluted, as a vapor,<sup>1-3</sup> or in a solvent such as ethyl acetate,<sup>4,5</sup> kerosene,<sup>6</sup> silicone fluid,<sup>7</sup> or benzyl alcohol.<sup>8</sup>

The solvent application procedure is reported to provide a more uniform coating on the glass surface. Improvements in the application of vapors and mists have also resulted in smoother coatings and more efficient use of the titanate esters.<sup>9-12</sup>

(TiO<sub>2</sub>)X films also are applied to glass or vitreous enamels for decorative purposes. Thin films enhance brilliance and thicker films impart an iridescent luster.<sup>13</sup> Aqueous solutions of chelated organic titanates may also be applied to hot or cold glass surfaces to impart scratch resistance.<sup>14-17</sup>

When the (TiO<sub>2</sub>)X film is 150 nm thick, it is antireflective to visible light yet reflective to heat-producing infrared radiation. Such coatings may be used to reduce specular reflection on window panes, camera lenses, binoculars, computer monitors, TV screens and other optical glass surfaces.<sup>18-21</sup> Heat-reflecting glass plates can be used as window panes or wall panels in buildings and vehicles to help prevent interior temperatures from rising.<sup>22,23</sup>

A (TiO<sub>2</sub>)X coating that promotes fiber adhesion to elastomeric and resinous materials can be obtained by treating a glass fiber with a solution of an organic titanate ester (e.g., Tyzor® TPT or TNBT), evaporating the solvent, hydrolyzing the resultant coating in moist air, and heating it to 75 to 200 °C (167 to 392 °F). The composites obtained by this procedure may be in the form of laminated sheets, molded objects, extruded products, etc.

If a glass fiber coated with a stabilized, aqueous solution of Tyzor® AA-65 is heated to 1700 to 2100 °C (3092 to 3812 °F), a glass ceramic fiber is obtained with improved modulus and temperature resistance.<sup>24</sup>

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